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PPMI Genetic Data Summary: Parkinson's Disease Risk SNPs and Polygenic Scores

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Summary

This data set includes genotype data from PPMI participant whole genome sequencing (WGS) for a pre-defined list of risk SNPs from Parkinson's disease GWAS results. This version includes SNPs and polygenic scores calculated with weights pulled from Nalls et al., 2019 [1], as well as SNPs from Hampton L. Leonard and Global Parkinson's Genetics Program (GP2), 2025 [2].

Method

WGS Data Sources: All available WGS was obtained from PPMI participants. Data from PPMI phase 1 (N=1825) was obtained from AMP-PD (https://amp-pd.org/) as pVCFs. Data from PPMI phase 2 was obtained from the Laboratory of Neuro Imaging Image and Data Archive (LONI; https://ida.loni.usc.edu) as gVCFs (N=1,544). In total, 3,369 PPMI participants had WGS available for analysis. Data was pulled using VCFtools [3] and BCFtools [4], and annotated with Annovar [5]. GBA p.N409S genotypes were checked against the PPMI PDGenes Consensus to verify match.

Risk SNP Selections and Polygenic Scores:

Nalls et al., 2019:

The list of 90 significant SNPs from the Nalls et al (2019) PD GWAS meta-analysis as well as the resulting SNP allele weights were obtained from https://github.com/GP2code/amp-pd-v3-pc-prs and https://github.com/neurogenetics/genetic-risk-score [1]. There were 88 of 90 SNPs present in the PPMI WGS. The two SNPs that were not available in the WGS data were rs6825004 and rs8087969, and proxy SNPs with $R^2 > 0.90$, matching the desired effect allele, and present in both phases were not found. Scores were calculated from the 88 SNPs and weights listed in **Table 1**.

Table 1. Nalls et al. 2019, 88 SNP list and beta weights

Chr	bp_hg38	dbSNP ID	Ref Alleles	Alt Allele	Beta weight
1	154925709	rs114138760	G	C	0.2812

1	155162560	rs35749011	G	A	0.6068
1	155235843	rs76763715	T	T	-0.7467
1	161499264	rs6658353	G	C	0.065
1	171750629	rs11578699	C	T	-0.0704
1	205754444	rs823118	C	T	0.1066
1	205768611	rs11557080	G	A	0.1315
1	226728377	rs4653767	T	T	0.0833
1	232528865	rs10797576	C	T	0.1114
2	17966582	rs76116224	A	A	0.1104
2	95335195	rs2042477	A	A	-0.0657
2	101780501	rs11683001	T	A	0.0705
2	134707046	rs57891859	A	A	0.0807
2	168253884	rs1474055	C	T	0.1796
3	18320267	rs73038319	A	A	-0.1693
3	28664199	rs6808178	T	T	0.0658
3	48711556	rs12497850	G	T	0.0636
3	122478045	rs55961674	C	T	0.0861
3	151391177	rs11707416	T	A	-0.0627
3	161359842	rs1450522	A	A	-0.0616
3	183042285	rs10513789	T	T	0.1485
4	931588	rs873786	C	T	-0.1731
4	958159	rs34311866	T	T	-0.2126
4	15735725	rs4698412	G	A	0.1035
4	17967188	rs34025766	T	A	-0.0839
4	76226816	rs4101061	A	A	-0.0912
4	76276901	rs6854006	C	T	-0.0912
4	89704960	rs356182	G	A	-0.2774
4	89715479	rs5019538	G	A	-0.1565
4	113447909	rs13117519	C	T	0.0875
4	169662006	rs62333164	G	A	-0.0638
5	60842132	rs1867598	A	A	-0.1554
5	103030090	rs26431	G	C	0.0621
5	134863415	rs11950533	C	A	-0.0916
6	27771022	rs4140646	G	A	0.0833
6	30140906	rs9261484	C	T	-0.0635
6	32610995	rs112485576	C	A	-0.1676
6	71778059	rs12528068	C	T	0.0657
6	111922088	rs997368	A	A	0.0714

6	132889222	rs75859381	T	T	-0.2207
7	23260430	rs199351	A	A	0.1016
7	66544864	rs76949143	T	A	-0.1432
8	11854934	rs1293298	A	A	0.093
8	16840084	rs620513	G	T	-0.0856
8	22668467	rs2280104	T	T	0.0556
8	129889663	rs2086641	T	T	-0.0605
9	17579692	rs13294100	T	T	-0.0859
9	17727067	rs10756907	A	A	-0.0926
9	34046393	rs6476434	C	T	-0.0615
10	15515407	rs896435	C	T	0.0735
10	102255522	rs10748818	A	A	-0.079
10	119656173	rs72840788	G	A	0.0763
10	119776815	rs117896735	G	A	0.4354
11	10537230	rs7938782	A	A	0.087
11	83776234	rs12283611	C	A	-0.0645
11	133917106	rs3802920	G	T	0.1073
12	40220632	rs76904798	C	T	0.1439
12	40340400	rs34637584	G	A	2.4289
12	46025303	rs7134559	C	T	-0.0539
12	122842051	rs10847864	G	T	0.1478
12	132487182	rs11610045	G	A	0.0601
13	49353596	rs9568188	T	T	0.0617
13	97212767	rs4771268	T	T	0.0675
14	37520065	rs12147950	T	T	-0.0529
14	54882151	rs11158026	C	T	-0.0842
14	74906331	rs3742785	A	A	0.0707
14	87997920	rs979812	G	T	0.061
15	61705186	rs2251086	T	T	-0.1186
16	19266171	rs6497339	A	A	0.063
16	28933075	rs2904880	C	C	-0.065
16	30966478	rs11150601	G	A	0.0907
16	50702745	rs6500328	A	A	0.0586
16	52602330	rs3104783	C	A	0.0668
16	52935514	rs10221156	G	A	-0.1156
17	7452302	rs12600861	A	A	-0.0565
17	42588995	rs12951632	T	T	0.0642
17	44216969	rs2269906	A	A	0.0631

17	44357262	rs850738	A	A	-0.071
17	45666837	rs62053943	C	T	-0.27
17	45720942	rs117615688	G	A	-0.2324
17	46789439	rs11658976	G	A	-0.0624
17	61840005	rs61169879	C	T	0.082
17	78429399	rs666463	A	A	0.076
18	33724354	rs1941685	G	T	0.0531
18	43093415	rs12456492	A	A	-0.0983
19	2341049	rs55818311	C	T	-0.0696
20	6025395	rs77351827	C	T	0.0802
21	37480059	rs2248244	G	A	0.0714

<u>Hampton L. Leonard and GP2, 2025:</u>
The list of 157 significant SNPs from a multi-ancestry Parkinson's disease (PD) GWAS meta-analysis as well as the resulting SNP allele weights were obtained from GP2, from the manuscript by Hampton L. Leonard and GP2 [2]. It should be noted that **there is some overlap between these data sets; however, there are also some SNPs unique to each set**. For the 157 SNPs, the beta weights from the 'all studies' category were used for polygenic score calculation. The polygenic score was calculated using the 157 SNPs and weights shown in **Table 2**.

Table 2. Hampton L. Leonard and GP2, 150 SNP list and beta weights

Chr	bp_hg38	dbSNP ID	Ref Alleles	Alt Allele	Beta weight
1	52724687	rs377808	T	G	-0.0486
1	112106855	rs12128848	A	G	-0.0414
1	155235843	rs76763715	T	C	0.8378
1	155236246	rs75548401	G	A	0.3649
1	155236376	rs2230288	C	T	0.4595
1	161000299	rs72712862	C	T	0.2103
1	161067049	rs72712893	C	G	0.2045
1	161327996	rs72714968	A	G	0.1536
1	161419856	rs61802091	G	A	-0.067
1	171747480	rs3049686	T	TAGAG	-0.0491
1	205770138	rs708723	C	T	0.0752
1	226731418	rs10495249	A	G	-0.0574
1	232508908	rs12073680	A	G	0.0672
1	243546888	rs7517340	T	C	-0.0506
2	17966582	rs76116224	A	T	-0.0731
2	23718478	rs12478701	T	C	0.0386
2	31572741	rs6749019	C	A	0.0397

2	32623787	rs13010404	T	G	-0.0531
2	61536072	rs9309337	T	C	-0.0516
2	69465874	rs9309428	C	T	0.0494
2	95326214	rs7563891	C	T	0.0458
2	101170053	rs60954733	A	G	-0.0455
2	134834675	rs6758044	T	C	-0.1014
2	147901026	rs3754541	G	A	-0.0546
2	157798690	rs10173157	A	G	-0.0506
2	160967491	rs1710669	C	G	0.0452
2	168129365	rs12618884	A	G	0.0398
2	168260515	rs2102808	G	T	0.1511
3	7024398	rs17046610	G	A	-0.1197
3	17710565	rs144678625	A	G	0.167
3	18247575	rs73034368	C	A	0.1309
3	28658687	rs1461806	A	G	-0.0503
3	39481512	rs1768208	T	C	-0.0418
3	58410136	rs56384862	A	G	0.0429
3	121842918	rs58088236	C	T	-0.0395
3	151405709	rs11717169	T	C	-0.0459
3	151946018	rs5853550	GAGAATTTC	G	-0.0539
3	151990484	rs10569457	CGAAA	C	-0.0532
3	155551064	rs359545	A	G	0.0573
3	161328149	rs11929238	A	T	0.0572
3	179199217	rs3729674	A	G	0.0622
3	183028420	rs34822404	C	T	-0.1167
4	940049	rs77186370	C	A	0.1209
4	950422	rs34884217	A	C	-0.1238
4	958159	rs34311866	T	C	0.1637
4	2934914	rs2857866	T	G	0.0721
4	15735725	rs4698412	G	A	0.072
4	18000605	rs59912548	ATC	A	-0.0974
4	76252761	rs13117238	C	A	0.0727
4	76278760	rs28628748	G	A	-0.0708
4	89704960	rs356182	G	A	-0.1875
4	89837794	rs2301134	A	G	0.041
4	113447909	rs13117519	C	T	0.0717
4	169711869	rs72696649	T	A	-0.052
4	169996341	rs75739600	A	C	0.0739

5	1431099	rs461753	C	T	-0.0449
5	60842132	rs1867598	A	G	0.1355
5	76303383	rs246814	C	T	0.084
5	80530472	rs10054911	T	A	-0.0867
5	124872055	rs11241774	A	T	0.0387
5	138443292	rs2350640	A	C	-0.0394
6	13664623	rs7741241	G	A	-0.0402
6	20023315	rs60740436	TA	T	0.0387
6	27224312	rs4713072	T	C	0.0462
6	27287064	rs6913724	A	T	0.0436
6	29015989	rs41316625	G	A	-0.0983
6	32603345	rs2647066	C	T	-0.1374
6	69763057	rs1236644794	AT	A	0.0562
6	71778059	rs12528068	C	T	0.0578
6	111836412	rs13340407	T	C	-0.0597
6	132828914	rs146617529	A	G	0.1992
7	23076996	rs6461688	G	A	-0.0775
7	95711872	rs2127183	A	C	-0.0411
7	100492237	rs6979335	T	C	-0.061
7	158605408	rs80148128	G	A	0.0388
8	1761106	rs10087654	G	A	0.0462
8	8457501	rs11277366	CCAGAGAGGCT	C	0.0422
8	11853604	rs34504234	CT	C	-0.0733
8	16840070	rs620490	T	G	-0.0824
8	22672162	rs4872005	G	C	-0.0425
8	109633256	rs6469273	C	G	0.0533
8	129873161	rs2663613	T	C	0.0392
9	17580618	rs8181077	A	T	-0.074
9	17701891	rs1536072	C	G	0.0786
9	34154849	rs13296299	T	A	0.0525
10	15507549	rs10796307	A	C	0.0464
10	102415138	rs2282295	G	A	0.0625
10	102875346	rs12764899	G	A	-0.0539
10	119651405	rs144814361	C	T	0.3234
10	119670121	rs2234962	T	C	0.0535
10	119856143	rs150084107	ACC	A	-0.066
11	83789062	rs12284617	A	G	-0.0416

11	113721110	rs10750027	G	A	-0.0402
11	133930880	rs11223628	G	A	0.0538
12	906303	rs2286028	G	C	0.0535
12	40006146	rs28370650	T	A	0.308
12	40179612	rs17443099	G	A	0.2697
12	40220632	rs76904798	C	T	0.11
12	40320097	rs35303786	T	C	0.1764
12	40340400	rs34637584	G	A	2.208
12	46022785	rs6582585	T	C	-0.0465
12	49549339	rs933738	A	G	0.0628
12	71785666	rs61754230	C	T	0.1756
12	100991017	rs7964712	A	G	-0.038
12	107466794	rs7350560	A	G	0.0464
12	109586736	rs2270375	A	G	-0.0429
12	122842051	rs10847864	G	T	0.1006
13	49341314	rs1198499	T	C	0.0467
13	97227087	rs1995052	T	C	-0.0414
13	109782884	rs1805097	C	T	0.0406
14	37749513	rs36003317	AATT	A	-0.0449
14	54881109	rs7155501	A	G	-0.078
14	87997920	rs979812	G	T	0.0439
14	100746019	rs201348288	CAAAACAA	C	0.0702
15	50701476	rs35336805	TAAAA	T	-0.0605
15	61701503	rs4774417	G	A	0.0733
15	64604181	rs648397	A	C	0.1556
15	88973692	rs34631560	G	A	-0.045
16	1666009	rs8058190	A	C	-0.0506
16	19266171	rs6497339	A	T	-0.0603
16	28893412	rs68040616	CAAAAAG	C	0.0423
16	31038712	rs732172	C	T	-0.0887
16	50811622	rs1861761	T	G	-0.0522
16	52610012	rs8046994	C	T	0.0564
16	52939375	rs1971534	A	G	0.0537
17	4932600	rs2243093	T	C	0.0585
17	7237287	rs222852	A	G	-0.0411
17	7594953	rs72827590	A	G	0.0584
17	16091574	rs2078050	C	T	-0.0448
17	17813217	rs12941356	A	G	0.0414

17	31296270	rs11080149	C	T	0.0641
17	42628829	rs12601457	C	T	-0.0507
17	44352876	rs5848	C	T	0.0604
17	44402945	rs34359486	G	A	0.0962
17	46239844	rs2532384	C	A	-0.2137
17	47535476	rs190825568	G	A	-0.0575
17	50146710	rs847680	T	C	-0.0526
17	61779284	rs11871753	A	G	0.0451
17	62016181	rs72843781	A	C	0.044
17	68281293	rs58426209	T	G	-0.0567
17	78425279	rs143392722	TTTATC	T	-0.0601
17	81996668	rs8074498	T	A	0.0429
18	34050285	rs35273482	GA	G	0.0653
18	43093415	rs12456492	A	G	0.0694
19	2353152	rs1128402	C	A	0.048
19	18482743	rs8105994	T	C	-0.044
19	32563836	rs11669800	G	A	0.1407
19	49145273	rs4802574	A	G	-0.0412
20	3172857	rs55785911	G	A	-0.0431
20	6005604	rs143230159	ACAAT	A	0.0625
20	31805133	rs35428411	ATGC	A	-0.0495
20	50535462	rs117517602	T	C	0.0692
21	37519947	rs11701836	A	G	0.064
21	40577012	rs4816690	T	C	0.0396
21	42221495	rs4148103	G	A	-0.0802
21	44366954	rs118075805	G	T	0.198
21	45161847	rs8130097	C	A	0.0971

Polygenic Score Calculations: Polygenic scores for each study were calculated using the PLINK v1.9 score function with the following code [6, 7]:

```
module load plink v1.9
plink --bfile $yourfile --score $scorefile --out $outputfilename
```

The **final data set** ('iusm_risk_snp_pgs') includes two tables: 'iusm_risk_snp_pgs_gt' and 'iusm_risk_snp_pgs.' The first of these is a genotype table and includes all SNPs in both data sets, with overlapping SNPs included in each SNP list only included once (**no duplicate SNPs**). Each SNP is listed in a separate column, with participant IDs as rows. Metadata for each SNP including chromosome, location in base pairs, and minor and alternate alleles, are included in the first six rows. Missing genotypes are represented by './.'. The second data set table includes calculated PGS. **Table 3** includes variable names and more information for each polygenic score. Scores calculated using Nalls et al., 2019 (90 SNP scores) are labeled "META5", and scores using SNPs from Leonard and GP2, 2025 (157 SNP scores) use the prefix "GP2". All eight scores are also accompanied by a corresponding column with the suffix "AC", showing the allele count, or number of SNPs used to calculate the score for each participant (which may differ if individual genotype data are missing).

Table 3. Polygenic Score Variables

Score Variable Name	Score SNP Count Variable Name	Source*	Description
GP2_PGS	GP2_AC	Leonard	All 157 SNPs processed
GP2_proxy_PGS	GP2_proxy_AC	Leonard	Five proxy SNPs were chosen for variants with high missingness; 157 SNPs processed
META5_PGS	META5_AC	Nalls	Nalls PGS files were used as-is; 88 SNPs processed out of 90
META5_excl_LRRK2_GBA_PGS	META5_excl_LRRK2_GBA_AC	Nalls	83/85 variants processed
META5_proxy_PGS	META5_proxy_AC	Nalls	88/90 SNPs processed; includes proxy SNPs selected in protocol
META5_proxy_excl_LRRK2_PGS	META5_proxy_excl_LRRK2_AC	Nalls	86/88 SNPs processed
META5_proxy_excl_GBA_PGS	META5_proxy_excl_GBA_AC_	Nalls	85/87 variants processed
META5_proxy_excl_LRRK2_GBA_PGS	META5_proxy_excl_LRRK2_GBA_AC	Nalls	83/85 variants processed

*'Nalls' indicates score is sourced from data presented in Nalls et al., 2019 [1]. 'Leonard' indicates score is sourced from data presented in Hampton L. Leonard and GP2, 2025 [2].

References

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